

UG0645
User Guide
Low Voltage Differential Signaling 7:1



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1 Introduction

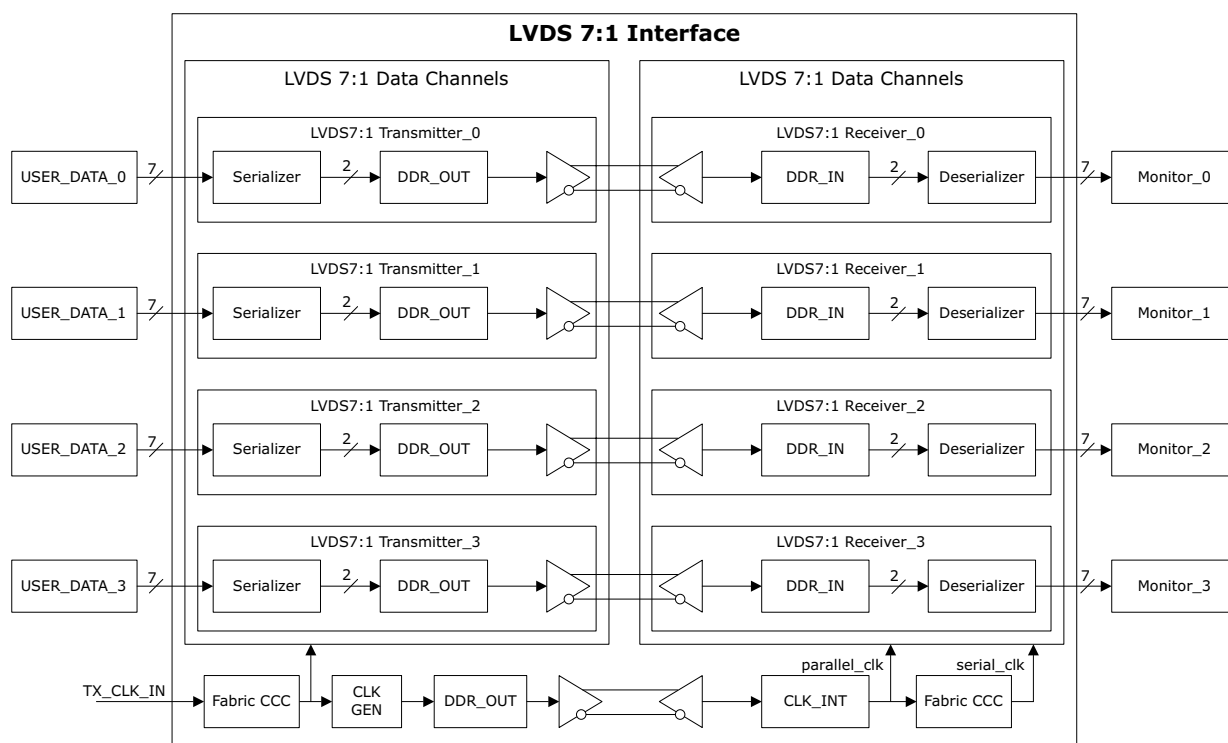
Low-voltage differential signaling (LVDS) is a high-speed, low-power, general-purpose interface standard. Also known as the ANSI/TIA/EIA-644 standard, LVDS was approved in March 1996. LVDS uses differential signaling with a nominal signal swing of 350 mV differential. The low signal swing decreases rise and fall times to achieve the maximum transmission rates specified in the LVDS standard. With LVDS, signal swing does not depend on the voltage of any specific supply.

LVDS uses current mode drivers, which limit power consumption. The differential signals are immune to ± 1 V common voltage noise. The Channel-Link technology was originally developed as a solution for flat panel displays, using LVDS for the physical layer (PHY). The technology was then extended into a method for general purpose data transmission. Channel-Link consists of a driver pair and a receiver pair. The driver accepts 28 single-ended data signals and a single-ended clock. The data is 7:1 serialized, and four data streams and a dedicated clock are driven over five LVDS pairs. The receiver accepts all four LVDS data streams and the LVDS clock, and then drives the 28 bits of data and the clock to the board.

Microsemi LVDS 7:1 is a source-synchronous interface that consists of multiple data bits and a clock. The LVDS 7:1 solution consists of five LVDS pairs: four data pairs and a clock pair. It provides four independent, source-synchronous channels, and on each channel, one cycle of parallel clock includes seven bits of serialized data. The serial data on the four channels together translates to 28 bits parallel data. It is recommended that the pinouts and clocks of all the four data channels be placed in the same I/O bank. However, the pinouts can be assigned to different banks, provided the clocking requirements for the design are met.

The following figure shows the top-level block diagram of the four-channel LVDS 7:1 interface.

Figure 1 • LVDS 7:1 Top-Level Block Diagram



The LVDS 7:1 transmit and receive blocks perform the following:

- The transmit block uses double data rate registers to transmit data on both the rising and falling edges of the clock. It multiplies the parallel clock by 3.5 and uses the clock to transmit seven serial bits of data in one parallel clock cycle.
- The receive block uses double data rate registers to capture data on both the rising and falling edge of the clock. It multiplies the captured clock by 3.5 to generate the input for the DDR_IN macro.

2 Hardware Implementation

2.1 Design Description

This section provides information on the implementation of the LVDS 7:1 receive and transmit modules.

2.1.1 LVDS 7:1 Receive Module

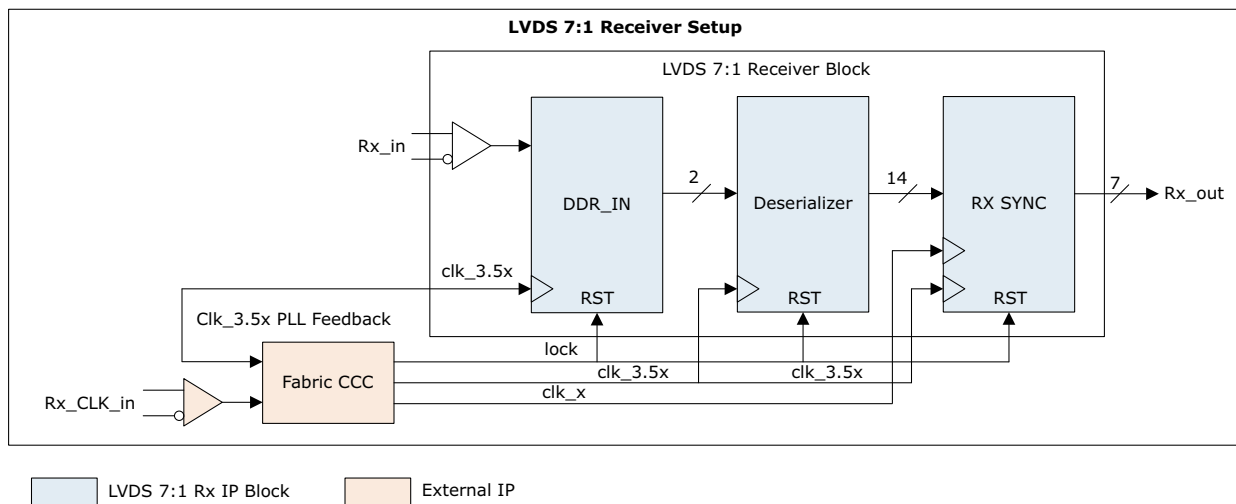
The LVDS 7:1 receive module receives LVDS data and an LVDS clock from the FPGA's high-speed LVDS buffers. The source-synchronous LVDS clock is passed to the fabric clock conditioning circuitry (CCC) block while the LVDS data is sent to the DDR_IN macro.

The fabric CCC block generates a serial clock (serial_clk) with a frequency of 3.5 times the input clock received from the transmitter (RX_CLK_IN).

The DDR_IN macro generates two streams of data at the rising edge of the serial clock. This data is then sent to the deserializer, which aligns the incoming data to a pre-defined training pattern. Once the data is aligned according to the word boundary defined in the training pattern, the align_serializer signal is enabled, and 14 bits of serialized data are sent to the RX_SYNC module. The RX_SYNC module synchronizes the 14-bit serial data to a parallel clock (parallel_clk), and then transmits a 7-bit parallel data as output.

The following block diagram shows the LVDS 7:1 receiver implementation, including input and output signals.

Figure 2 • LVDS 7:1 Receiver Block Diagram



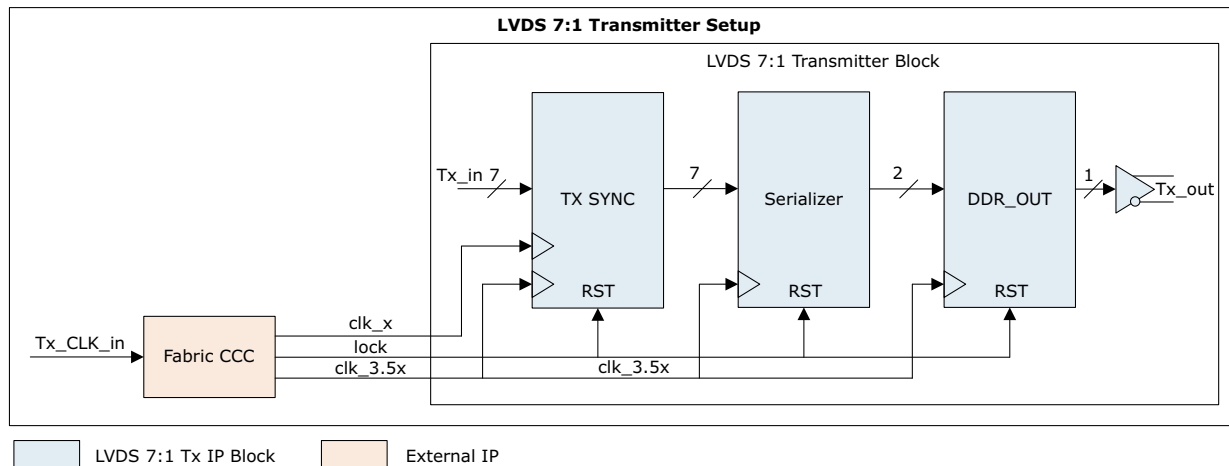
2.1.2 LVDS 7:1 Transmit Module

The LVDS 7:1 transmit module receives seven bits of parallel data and a parallel clock, TX_CLK_in. TX_CLK_in is sent to the fabric CCC block, which generates two output clocks: a parallel clock with the same frequency as TX_CLK_in, and a serial clock with a frequency 3.5x that of TX_CLK_in.

The TX_SYNC module synchronizes the incoming seven bits of parallel data with the output clocks and transmits the synchronized data to the serializer block. The serializer block then serializes the seven bits of parallel data into two bits of serial data. These two bits of serial data are passed to the DDR_OUT macro, which operates at the serial clock's frequency. The serialized output data of the DDR_OUT macro is then transmitted out of the device using high-speed LVDS buffers.

The following block diagram shows the LVDS 7:1 transmitter implementation, including input and output signals.

Figure 3 • LVDS 7:1 Transmitter Block Diagram



2.2 Inputs and Outputs

The following table lists the LVDS 7:1 receiver interface input and output ports.

Table 1 • LVDS 7:1 Receiver Interface Ports

Signal Name	Direction	Width (in bits)	Description
Reset_n	Input	—	System reset
Serial_Clk	Input	—	System serial clock
parallel_clk	Input	—	Parallel clock
CAM_D3_P	Input	—	Serial input channel D rising
CAM_D3_N	Input	—	Serial input channel D falling
CAM_D2_P	Input	—	Serial input channel C rising
CAM_D2_N	Input	—	Serial input channel C falling
CAM_D1_P	Input	—	Serial input channel B rising
CAM_D1_N	Input	—	Serial input channel B falling
CAM_D0_P	Input	—	Serial input channel A rising
CAM_D0_N	Input	—	Serial input channel A falling
RDATA_A	Output	7	Parallel output data channel A
RDATA_B	Output	7	Parallel output data channel B
RDATA_C	Output	7	Parallel output data channel C
RDATA_D	Output	7	Parallel output data channel D
Align_serializer_a	Output	—	Channel A Rx alignment signal to the training pattern
Align_serializer_b	Output	—	Channel B Rx alignment signal to the training pattern
Align_serializer_c	Output	—	Channel C Rx alignment signal to the training pattern

Table 1 • LVDS 7:1 Receiver Interface Ports

Signal Name	Direction	Width (in bits)	Description
Align_serializer_d	Output	–	Channel D Rx alignment signal to the training pattern
Training pattern	Input	–	Pattern used to align Rx to Tx
CAM_CLKOUT_N	Input	–	Clock from fabric
CAM_CLKOUT_P	Input	–	Clock from fabric
Rclk_o	Input	–	LVDS receiver clock output

The following table lists the LVDS 7:1 transmit interface input and output ports.

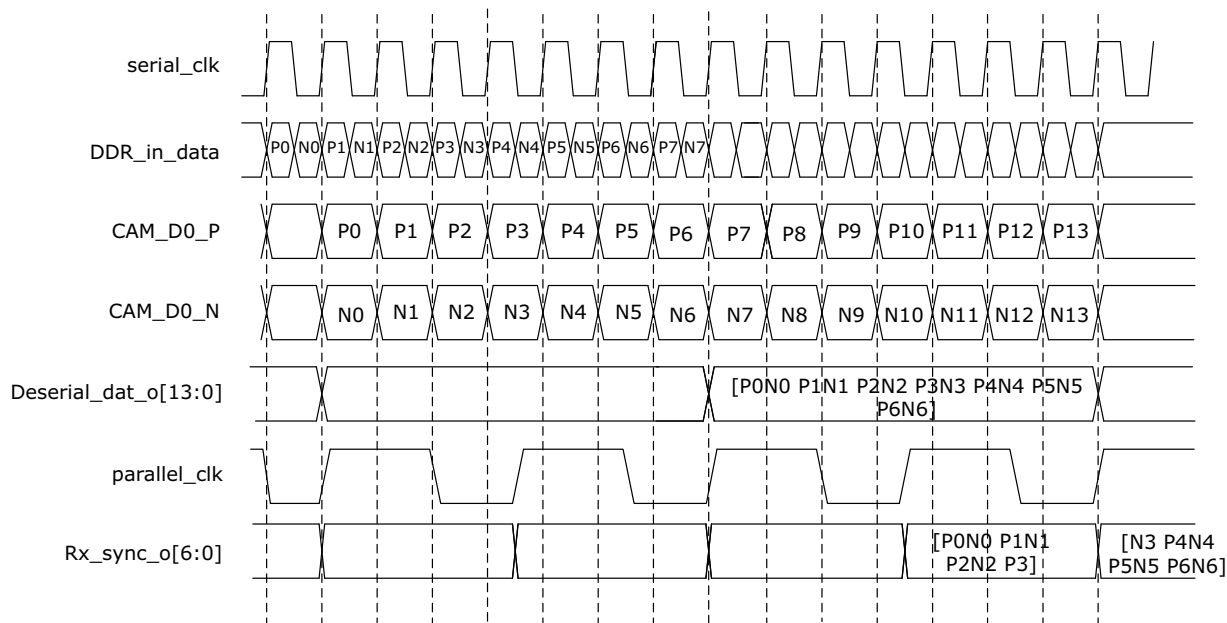
Table 2 • LVDS 7:1 Transmitter Interface Ports

Signal Name	Direction	Width (in bits)	Description
RESET	Input	–	System reset
serial_clock	Input	–	System serial clock
parallel_clk	Input	–	Parallel clock
WDATA_A	Input	7	parallel data channel A
WDATA_B	Input	7	Parallel data channel B
WDATA_C	Input	7	Parallel data channel C
WDATA_D	Input	7	Parallel data channel D
PADP_TX_0	Output	–	Serial channel A rising data
PADN_TX_0	Output	–	Serial channel A falling data
PADP_TX_1	Output	–	Serial channel B rising data
PADN_TX_1	Output	–	Serial channel B falling data
PADP_TX_2	Output	–	Serial channel C rising data
PADN_TX_2	Output	–	Serial channel C falling data
PADP_TX_3	Output	–	Serial channel D rising data
PADN_TX_3	Output	–	Serial channel D falling data
PADN_CLK_OUT	Output	–	Output differential negative TX clock
PADP_CLK_OUT	Output	–	Output differential positive TX clock

2.3 Timing Diagrams

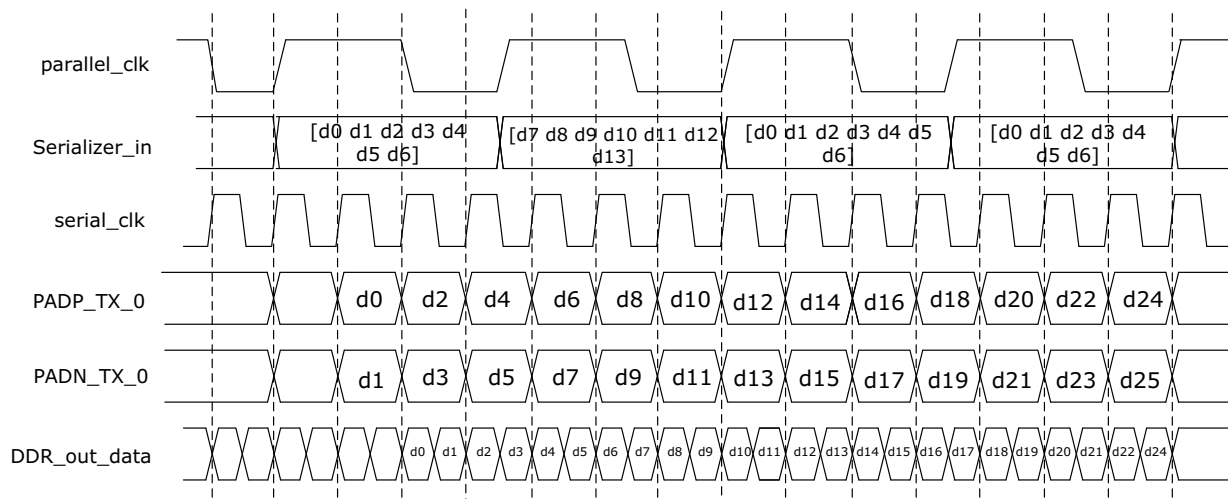
The following figure shows the timing diagram for the LVDS 7:1 receiver.

Figure 4 • LVDS 7:1 Receiver Timing Diagram



The following figure shows the timing diagram for the LVDS 7:1 transmitter.

Figure 5 • LVDS 7:1 Transmitter Timing Diagram



2.4 Loopback Test

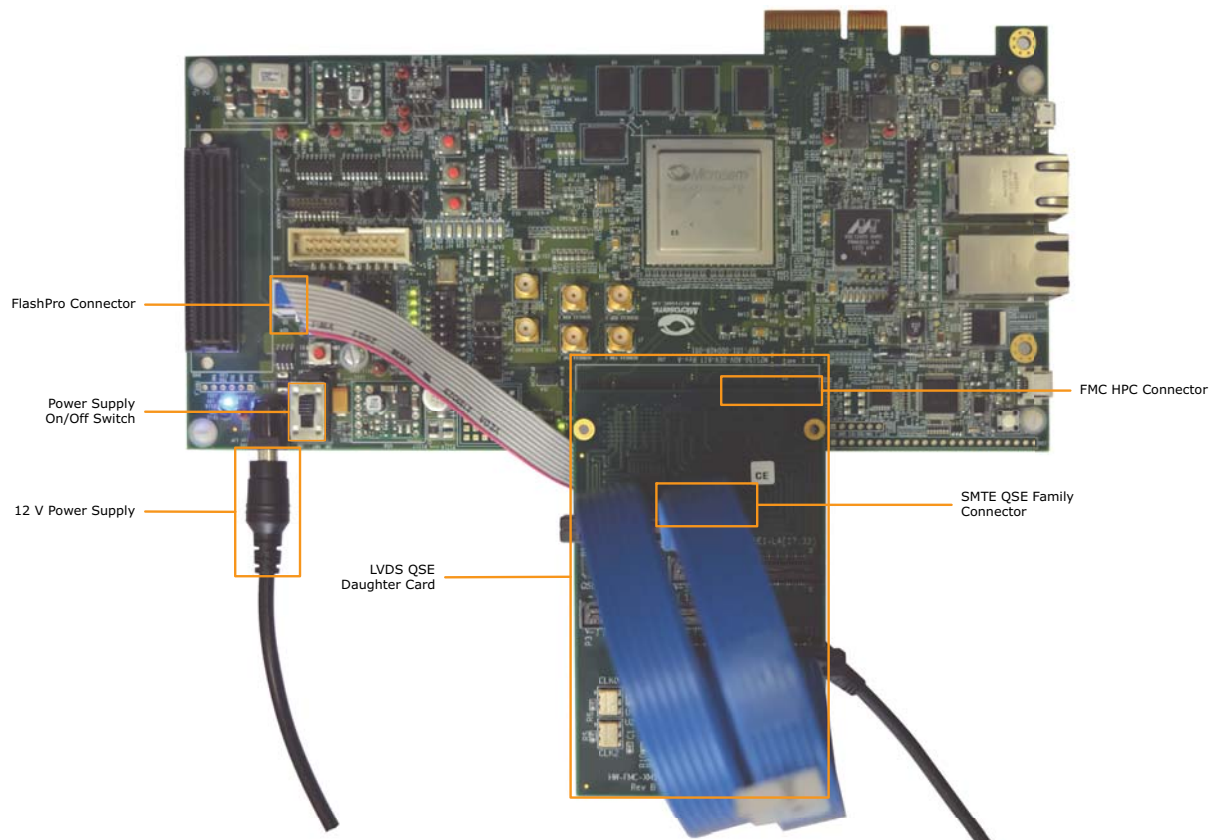
A loopback test performed on the Microsemi SmartFusion2® M2S150 Advanced Development Kit to validate the LVDS 7:1 functionality. The test setup consisted of five parallel transmit and receive channels (four data channels and one clock channel), and the test was performed at varying speeds up to 90 MHz.

The M2S150 Advanced Development Kit provides an FMC high pin count (HPC) J64 connector interface. For the test, an LVDS QSE daughter card was installed on the HPC connector. This daughter card provides four Samtec QSE family connectors, which break out as a subset of the FMC HPC signal set.

The FMC HPC signals were connected as length-matched pairs to the QSE connectors. The pairs were then loop-backed using the Samtec loopback cable.

The following figure shows the test setup for the LVDS 7:1 loopback test.

Figure 6 • LVDS 7:1 Loopback Test Setup



2.5 Performance Statistics

During the LVDS 7:1 interface design validation performed using M2S150 Advanced Development Kit, it was found that the LVDS 7:1 interface design successfully operated at a maximum frequency of 90 MHz for the parallel clock and 315 MHz for the serial clock with 3.5x the parallel clock's frequency. Overall data transfer rates of up to 630 Mbps of data were achieved per LVDS pair. For more information about the test setup used for the validation, see [Loopback Test](#), page 6.

2.6 Resource Utilization

The following table shows the resource utilization of a sample LVDS 7:1 block implemented in a SmartFusion2 M2S150T-1152FC device.

Table 3 • LVDS 7:1 Receiver Resource Utilization

Resource	Usage
D flip-flops (DFF)	990
4-input look-up tables (LUT)	590
Math blocks (MACC)	0
RAM1Kx18	0
RAM64x18	4

Table 4 • LVDS 7:1 Transmitter Resource Utilization

Resource	Usage
DFFs	190
4-input LUTs	80
MACC	0
RAM1Kx18	0
RAM64x18	0

3 Revision History

The revision history describes the changes that were implemented in the document. The changes are listed by revision, starting with the most current publication.

3.1 Revision 3.0

The following is a summary of the changes in revision 3.0 of this document:

- Information about LVDS 7:1 loopback test was added. For more information, see [Loopback Test](#), page 6.
- Performance statistics were updated. For more information, see [Performance Statistics](#), page 7.
- Resource utilization details were updated. For more information, see [Resource Utilization](#), page 8.

3.2 Revision 2.0

The following is a summary of the changes in revision 2.0 of this document.

- Updated the SAR (76159).

3.3 Revision 1.0

Revision 1.0 was the first publication of this document.